LABORATORY MANUAL

SC2207/CZ2007: Introduction to Databases

Implementation of a Database Application

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

NANYANG TECHNOLOGICAL UNIVERSITY

1. <u>OBJECTIVES</u>

Upon completion of the assignment, the student should be able to:

a. Construct an entity-relationship model at a conceptual level.
b. Map the model into a schema of a relational DBMS.
c. Implement the given schema on a relational DBMS.
d. Use a database language (SQL) to retrieval data from a relational DBMS.

2. <u>INSTRUCTIONS</u>

a) **<u>Team Formation</u>**: This is a team-based assignment. Each team consists of **five to seven** members from your laboratory group, to be approved by your lab supervisor, to be formed during the Lab 1 session. The lab supervisor may add or remove members from your teams to ensure an even spread and mix of students in each team. The final members of your team must be submitted to the lab executive during the **Lab 1 session**.

b) **<u>Lab Submission</u>**: There are five scheduled lab sessions for this team assignment. Laboratory sessions will start from **Week 3** for those scheduled on odd weeks, or **Week 4** for those scheduled on even weeks. For Lab 1, 3 or 5 submissions, do include a **cover page** indicating the team number and team members. Names of team members must appear as they do in student cards; do not shorten or use nicknames or aliases. There is a penalty of 5 marks per day of late submission.

c) **<u>Lab Attendance</u>**: Attendance is taken for the **first**, **third** and **fifth** lab sessions only. Attendance for the second and fourth lab sessions is not mandatory.

d) **<u>Fair Participation</u>**: Each student is expected to make fair and equal contribution to EACH lab, and thoroughly understand the expectations of EACH lab. It should NOT be the case that a student contributes less to Lab X and make up for it by contributing more to Lab Y. Each submission needs to indicate contributions from each member. The final marks of a team member may be adjusted based on the team score and individual contribution. **Appendix C is to be submitted with each submission**.

e) **<u>Lab Supervisors</u>**: For each lab session, there is a lab supervisor and a lab executive assisting you. The lab supervisor is a professor or a teaching assistant whom you may approach for clarifications on lab work, lab report submission, graded lab reports, etc. The lab executive is a technical staff whom you may approach for lab logistics (lab attendance, SQL Server account matter, lab submission deadline, computer problem, etc.).

f) **<u>AI tools</u>**: NTU's policy on the use of AI tools: "The University requires students to (i) identify any generative AI tools used and (ii) declare how the tools are used in submitted work. Please note that even with acknowledgement, copying of output generated by AI tools (in part or whole) **may still be regarded as plagiarism**." **Appendix D is to be submitted with each submission**.

3. INTRODUCTION

The assignment covers the portion of the course concerning data modelling, database design and implementation from the user's viewpoint. Thus, the assignment involves modelling as well as implementation aspects of the database course.

The overall aim of the laboratory is to develop an application based on a given data model using a given database management system. This exercise will bring you through a crucial first part of the life cycle of a database application. It is assumed that the data analysis has been performed. Note that this manual provides you with more information than is required for the first laboratory session, e.g., not all constraints can be modelled in the beginning but are included at a later implementation stage. In contrast you might require additional information for an understanding of the application. Proceed by stating your assumptions in written form and / or ask your laboratory supervisor.

4. DESCRIPTION OF THE ASSIGNMENT

The description of the application is given in Appendix A and B. This includes background and general requirements of the application, conceptual information about the system and its users as well as a list of SQL queries that must be fulfilled as a minimum. Note that teamwork is required. Each team will submit one solution. **No individual submission will be accepted.**

4.1 First Laboratory Session: Creating an ER Diagram

Appendix A gives conceptual information about the project obtained after a partial system analysis was performed. Based on the description, construct a suitable ER diagram. Analyze the choice of entity sets, different types of relationships required, the usage of weak entity sets, subclasses, etc. and compare them with alternative solutions from your team members. You need to submit the following, latest **three working days** after the first laboratory session, to the NTULearn course site for your lab group:

- A PDF document of your ER diagram. A good ER diagram is one that is self-explanatory. If you believe certain parts of your ER diagram need explanation, you can include a written description (maximum one page). Combine both the ER diagram and the written explanation (if any) as a single PDF document, labeled as follows: Lab1_XXX_TeamY.pdf, where XXX is your lab group and Y is your team number. Marks are given for neat presentation of your ER diagram.

- Assessment for Lab 1 is based on whether the submitted ER diagram reflects correct understanding of ER diagram artefacts (entity sets, relationships, weak entities, subclasses, etc.) and whether they are used correctly and appropriately. Do note that not all information given in Appendix A can be represented in an ER diagram and that more than one ER diagrams are possible. It is part of the project work for your team to submit what your team deems to be the best ER diagram among all possible ones.

4.2 Second Laboratory Session: Finalization of the ER Diagram

There is no submission for the second laboratory session. In this lab, each team should finalize their database design based on the feedback received from their lab supervisor and prepare for Lab 3. Please note that the second laboratory session is a free access session, i.e., attendance is not mandatory.

4.3 Third Laboratory Session: Generation of Normalized Database Schema

In this lab, you convert the ER diagram into relational schema and ensure that the relations are at least in 3NF. Follow the general guidelines covered during the lectures and tutorials to produce suitably normalized relations. For each relation, the key(s), primary key, and functional dependencies must be specified. If a relation is generated due to the normalization of an original relation, then the normalization steps must be presented. You need to submit the following, latest **three working days** after the third laboratory session, to the NTULearn course site for your lab group:

- A PDF document of the normalized database schema and FDs associated with each relation. Label the PDF document as: Lab3_XXX_TeamY.pdf, where XXX is your lab group and Y is your team number. If a relation that is created from the ER diagram violates 3NF, then this should be highlighted along with the decomposed normalized relations. Note that for this lab, no SQL code should be submitted. Hence, the structure of your solution shall be similar to the following example:

  R1(A, B, C, D)
  Keys: AB, AD
  Primary Key: AB
  FDs: AB → CD, A → D
  The relation is in 3NF. (If relation is not in 3NF, perform the steps of the 3NF normalization.)

- Assessment for Lab 3 is based on whether the submitted report reflects correct understanding of keys in relations, identification of appropriate functional dependencies in each relation, how normalized relations are formed, and whether the normalizations are correctly and appropriately performed. Do note that in your final set of relations, the keys and functional dependencies in each relation may not be explicitly given in the description in Appendix A.

4.4 Fourth Laboratory Session: Implementation of the database schema

There is no submission for the fourth laboratory session. In this lab, the finalized database schema is to be implemented using SQL DDL commands. Your tables should be appropriately populated with sufficiently realistic records using SQL INSERT statements so that your query solution for Appendix B results in some meaningful output records (3 to 5) for each query. Your implementation should clearly incorporate the primary and foreign keys, data types, and any form of constraints. The lab provides MS SQL Server software for your implementation. You should start to work on the queries in Appendix B.

Please note that the fourth laboratory session is a free access session, i.e., attendance is not mandatory.

4.5 Fifth Laboratory Session: Final demonstration

In this lab, the implementation obtained from the previous laboratory session must now be extended to provide SQL query solutions for the queries in Appendix B. **At the end of the lab session**, you need to submit a single PDF document to the NTULearn course site for your lab group containing the followings:

- SQL DDL commands for table creation (from Lab 4).
- SQL statements to solve the queries in Appendix B and additional queries (if any). Each query should be immediately followed by the query output. Briefly explain how the output is obtained.
- A printout of all table records.
- Description of any additional effort made.

Label the PDF document as: Lab5_XXX_TeamY.pdf, where XXX is your lab group and Y is your team number. You should prepare the PDF document in advance before coming to the lab. Some DDL commands may look like this:

```
CREATE TABLE name (
        attr1 datatype NOT NULL,
        attr2 datatype,
        ...
        PRIMARY KEY (attr1),
        FOREIGN KEY (attr3) REFERENCES name(attr1)
                ON DELETE ... ON UPDATE ...,
```

);

In addition to the PDF document, you are to capture **screen recording** of query execution as a mp4 video file. For each query in Appendix B and additional queries, first show the SQL statement, then execute the query and show the query results, all recorded as a mp4 video file. Each query video should be no more than 30 seconds and labeled as: Lab5_XXX_TeamY_Q#.mp4 where # is the query number. Zip the PDF and all mp4 files into one single ZIP file.

During the lab session, you may be given additional queries to solve. In addition, your lab supervisor may require in-person live demonstration and Q&A. All team members are to actively contribute during the demonstration session and be familiar with **all aspects** of the project. No slide presentation is required.

APPENDIX A: APPLICATION DESCRIPTION

GlobalLogistics is a comprehensive supply chain management company that handles warehousing, transportation, inventory management, and order fulfillment for e-commerce and retail businesses worldwide. You have been asked to design the database system for their operations.

1. Client Companies: GlobalLogistics serves various client companies (merchants) who outsource their logistics operations. Each client has a unique client ID, company name, primary contact person, contract start date, and service tier (bronze, silver, gold, platinum). Clients may operate in multiple countries and require region-specific logistics solutions.

2. Products: Client companies register their products in the system. Each product has a unique ID, product name, category, brand, unit cost, and retail price. Products have physical attributes including weight, dimensions (length, width, height), and special handling requirements (fragile, hazardous, temperature-controlled, high-value).

3. Warehouses & Storage Facilities: GlobalLogistics operates multiple warehouses globally. Each warehouse has a warehouse code, address, total warehouse size, temperature control capabilities (ambient, refrigerated, frozen), and security level. Warehouses are organized into storage zones (receiving, bulk storage, picking area, packing area, shipping dock), and each zone contains multiple storage locations with location codes.

4. Inventory Management: Products are stored in warehouses, and the system tracks inventory levels in real-time. For each product-warehouse combination, the system records quantity on hand, quantity reserved (for pending orders), quantity available for sale, reorder quantity, and location within the warehouse. Inventory movements (receipts, putaways, picks, adjustments) are logged with timestamps and reasons. Client and warehouse generally sign a contract that is valid for a certain period whereby products from the client will be stored in the warehouse.

5. Suppliers: Client companies source products from suppliers/manufacturers. Each supplier has a supplier ID, company name, country of origin, payment terms, and lead time. Suppliers may provide multiple products to multiple clients, and the system tracks which supplier can provide which products at what cost and lead time. Supplier and client generally sign a contract that is valid for a certain period whereby the supplier will supply product(s) to the client.

6. Purchase Orders: When client inventory runs low, purchase orders are generated to suppliers. Multiple clients may consolidate and generate one single PO. Each PO has a PO number, order date, supplier(s), delivery warehouse, order status (draft, submitted, confirmed, partially received, fully received, cancelled), and total order value. A PO contains multiple line items, each specifying a product, ordered quantity, unit price, and expected delivery date.

7. Inbound Shipments: When suppliers ship products to warehouses, inbound shipments are created. Each shipment has a shipment ID, PO reference, origin location, destination warehouse(s), tracking number,

shipped date, expected arrival date, and actual arrival date (in transit, arrived, receiving in progress, completed). A shipment contains multiple line items, each specifying a product, shipped quantity, and expected arrival date. Upon arrival, warehouse staff receive shipments by scanning products and verifying quantities against PO line items.

8. Vehicles & Fleet Management: For last-mile delivery in certain regions, GlobalLogistics operates its own delivery fleet. Each vehicle has a vehicle ID, type (van, truck, refrigerated truck), license plate, capacity (weight and volume), and assigned driver. Vehicles are assigned to delivery routes daily.

9. Drivers & Warehouse Staff: GlobalLogistics employs drivers and warehouse personnel. Each employee has an employee ID, name, employment type (full-time, part-time, contract), hire date, assigned facility or region, and certifications (forklift operator, hazmat handling, CDL license class for drivers). Drivers specifically record their license number, and license expiration date.

10. Delivery Routes: For fleet-operated deliveries, daily delivery routes are planned and optimized. Each route has a route ID, assigned driver, assigned vehicle, route date, total distance, total stops, and route status (planned, in progress, completed). A route contains multiple stops, each associated with an order delivery or pickup, with planned sequence, estimated arrival time, and actual arrival time.

11. GlobalLogistics uses artificial intelligence to automatically create optimized schedules for the delivery fleet given the current availability of drivers, the current availability of the different types of vehicles, the amount of products to be delivered and their required dates of arrivale, and the locations of the warehouses. The use of AI will give the company a competitive advantage over their competitors.

12. Your database should support the queries listed in Appendix B.

Note that the information above may not be complete. Some aspects of the database application's details may have been omitted. It is expected that you come up with your own solution(s) in case of inconsistencies or missing information. However, you have to keep track of these aspects and explain your assumptions in your submitted report.

APPENDIX B: QUERIES

1. For each warehouse, find its' top three clients (those who brought in the most amount of business in dollar terms to the warehouse).
2. Do warehouses located in Singapore have more businesses (in dollar terms) than warehouses in Los Angeles, USA?
3. What are the top three months in a year for the last two years that have the most purchase orders created in the system?
4. What is the average length of time (in terms of months) it takes for products ordered until the products are delivered to warehouses?
5. Which suppliers only supply products to warehouses in Singapore?
6. Which suppliers do not supply any product to warehouses in Thailand but have supplied all the products in warehouses in Singapore?
7. Which are the locations where shipment depart from that have experienced the most delays (actual arrival date is more than 6 months after expected arrival date)?

Note: For all the above queries, you must populate your tables with sufficient data to generate concrete result outputs. No query should generate a NIL (empty) output.

APPENDIX C: INDIVIDUAL CONTRIBUTION FORM

Full Name	Individual Contribution to Lab 1 Submission	Percentage of Contribution	Signature

Full Name	Individual Contribution to Lab 3 Submission	Percentage of Contribution	Signature

Full Name	Individual Contribution to Lab 5 Submission	Percentage of Contribution	Signature

APPENDIX D: USE OF AI TOOL(S) IN LAB WORK

Each team member should indicate either A or B:

A. I affirm that my contribution(s) to the lab work is my own, produced without help from any AI tool(s).
B. I affirm that my contribution(s) to the lab work has been produced with the use of AI tool(s).

Team member (full name)	Signature	Date	A or B

By signing this form, you declare that the above affirmation made is true and that you have read and understood NTU's policy on the use of AI tools.

If any team member answered B, the team member(s) must indicate and replicate the table below for every instance that AI tool(s) is used:

Name of AI tool	< For example, ChatGPT >
Input prompt	< Insert the question that you asked ChatGPT >
Date generated	
Output generated	< Insert the response verbatim from ChatGPT >
Output screenshots	
Impact on submission	< Briefly explain which part of your submitted work was ChatGPT's response applied >